Lecture 5 // Sept 8 // Yang Mills Functional & Equations

Recall

We defined the following objects: (M^4, g), E an SU(2) bundle over M.

$$\mathcal{A} := \text{space of SU(2) connections } \{\nabla = d + A\}, \text{ this is an affine space over } \Gamma(T^*M \otimes \operatorname{ad} E).$$

$$\mathcal{G} = \text{SU(2) gauge group} = \Gamma(\operatorname{Aut} E), \qquad g \curvearrowright A, \qquad A \mapsto g^{-1}Ag + g^{-1}dg.$$

$\mathcal{M}$ = Moduli space of Yang Mills connections with self dual curvature, i.e.,

$$\mathcal{M} = \{\text{SU(2) connections } \nabla = d + A \text{ such that } F_A = \star F_A \text{ (self dual)}\}/\mathcal{G}.$$

This condition of having self-dual curvature is actually a specific instance of a larger phenomena, which we'll discuss today to put A(SD) connections in context.

The Yang Mills Functional

Recall that the Lie algebra of SU(2), $\mathfrak{su}(2)$, is defined as the subset of 2×2 complex matrices which are trace-free and skew-hermitian.

$$\mathfrak{su}(2) = \left\{ \begin{pmatrix} ix & -y + iz \\ y + iz & -ix \end{pmatrix} \;\middle|\; (x, y, z) \in \mathbb{R}^3 \right\}$$

In particular, this definition clearly shows that SU(2) is in bijection with $\mathbb{R}^3$. $\mathfrak{su}(2)$ can be endowed with an inner product given by $\langle A, B \rangle = \operatorname{tr}(AB^\dagger) = -\operatorname{tr}(AB)$, which is the same inner product as the one on $\mathbb{R}^3$ (the dot product).

$$\left\langle \begin{pmatrix} ix & -y + iz \\ y + iz & -ix \end{pmatrix}, \begin{pmatrix} ip & -q + ir \\ q + ir & -ip \end{pmatrix} \right\rangle = 2(xp + yq + zr) = 2 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

This inner product on $\mathfrak{su}(2)$ can be extended to an L^2 inner product on $\Omega^i(\operatorname{ad} E) = \Gamma(\Lambda^i T^*M \otimes \operatorname{ad} E)$. Given two sections F and G, their inner product is defined as

$$\langle F, G \rangle_{L^2} = -\int_M \operatorname{tr}(F \wedge \star G)$$

where the inner product and trace is taken pointwise for each $p \in M$, giving a 4-form at each point, i.e., a function of p which we can integrate over M. By taking $G = F$, the norm of F is thus $\|F\|^2 = \langle F, F \rangle_{L^2}$.

Using this inner product, we may now define the Yang-Mills Functional.

Definition 1 (Yang-Mills Functional)

$$\mathrm{YM} : \{\mathrm{SU}(2)\text{-connections}\} \to \mathbb{R}$$
$$\nabla \longmapsto \mathrm{YM}(\nabla) := \langle F_A, F_A \rangle_{L^2} = -\int_M \mathrm{tr}(F_A \wedge \star F_A) = -\int_M \|F_A\|^2 \, d\mathrm{vol}_M$$

Heuristically, $\mathrm{YM}(\nabla)$ measures the "strength" of curvature of ∇. For some connection, what does it mean to be a critical point of the Yang-Mills functional? Recall that the space of connections is affine over $\Gamma(T^*M \otimes \mathrm{ad}\, E)$. A fixed connection ∇ is a critical point of YM if every nearby connection ∇' has larger or smaller curvature. Any such ∇' can be written as $\nabla' = \nabla + tB$, i.e., a small perturbation of ∇ by some section $B \in \Gamma(T^*M \otimes \mathrm{ad}\, E)$.

In terms of F_A, the curvature of ∇' becomes

$$F_{A+tB} = d(A + tB) + (A + tB) \wedge (A + tB) = F_A + t(dB + A \wedge B + B \wedge A) + t^2 B \wedge B.$$

Define $D_A B := dB + A \wedge B + B \wedge A$.

The derivative $\left.\frac{d}{dt}\right|_{t=0} \mathrm{YM}(\nabla + tB)$ then becomes

$$\left.\frac{d}{dt}\right|_{t=0} \mathrm{YM}(\nabla + tB) = \left.\frac{d}{dt}\right|_{t=0} \langle F_{A+tB}, F_{A+tB} \rangle$$

$$= \left.\frac{d}{dt}\right|_{t=0} \langle F_A + tD_A B + t^2 B \wedge B,\ F_A + tD_A B + t^2 B \wedge B \rangle$$

$$= 2\langle D_A B, F_A \rangle.$$

Therefore ∇ is a critical point of YM iff this pairing is 0 for all B.

Before we continue, we should discuss the appearance of $D_A B$.

Interlude: More about connections & bundles

Recall the usual deRham complex,

$$\Omega^0 \xrightarrow{d} \Omega^1 \xrightarrow{d} \Omega^2 \xrightarrow{d} \cdots$$

Now, take a vector bundle $E \to B$ and a fixed connection ∇ on E. We have already described ∇ as a map $\nabla : \Gamma(E) \to \Gamma(T^*M \otimes E)$. Can we get an analogous complex to the deRham complex?

To do so, we must generalize the spaces & the maps. Recall that the Ω^i in the deRham complex are defined to be sections of the bundle $\Lambda^i T^*E \to M$.

Definition 2

E-valued i-forms are sections of the bundle $\Lambda^i T^*M \otimes E$:

$$\Omega^i(E) = \Gamma(\Lambda^i T^*M \otimes E).$$

Note that when $i = 0$, we recover $\Omega^0(E) = \Gamma(\Lambda^0 T^*M \otimes E) = \Gamma(E)$, and when $i = 1$, we get

$\Omega^1(E) = \Gamma(\Lambda^1 T^*M \otimes E)$. So as defined above, a connection gives a map $\nabla : \Omega^0(E) \to \Omega^1(E)$. We extend the definition of ∇ using the Leibniz rule, as follows.

Definition 3

$$\nabla : \Omega^i(E) \to \Omega^{i+1}(E)$$

$$\nabla(\alpha \otimes s) = d\alpha \otimes s + (-1)^i \alpha \wedge \nabla s$$

where α is an i-form and s a section of E, so $(\alpha \otimes s) \in \Omega^i(E)$.

Note that unlike in the deRham complex, $\nabla^2 \neq 0$. However, measuring the failure of this fact is very useful.

Lemma 4
$\nabla^2(s) = F_A s$.

Proof. Choose basis sections $\{e_1, \ldots, e_n\}$ by choosing a trivialization of the bundle. Then $s = \sum s_i e_i$, where s_i are functions.

Recall $\nabla = d + A$. Therefore

$$\nabla^2 s = \nabla(\nabla s) = \nabla \left(\sum_{i,j} ds_i \otimes e_i + s_i A_{ji} e_j \right) \quad \text{(Einstein but we have a single sum here)}$$

$$= \underbrace{-ds_i \wedge (s_i A_{ji} e_i)}_{\nabla \text{ of 1st term}} + \underbrace{\nabla(s_i A_{ji}) e_j \wedge ds_i e_i}_{\nabla \text{ of 2nd term}}$$

$$= \underbrace{-ds_i \wedge (s_i A_{ji} e_i)}_{\nabla \text{ of 1st term}} + \big(ds_i \wedge A_{ji} + s_i dA_{ji}\big) e_j - (s_i A_{ji}) \wedge (A_{kj} e_k)$$

$$= s_i \, dA_{ji} e_j + s_i A_{j\ell} A_{\ell i} e_j$$

$$= dA + A \wedge A. \qquad \square$$

The upshot here is that curvature is the obstruction for the differential on the $\Omega^i(E)$ complex to square to 0.

We'll now consider the bundle $\mathrm{End}(E)$. Given a connection $\nabla = d + A$ on E, we can define a connection on $\mathrm{End}(E)$ using ∇. We do so locally.

Definition 5
A connection D_A on $\mathrm{End}(E)$ is locally of the form

$$D_A B := dB + AB - BA$$

where A is a matrix of 1-forms and B is a section of $\mathrm{End}(E)$, i.e., a matrix valued function.

This definition can be motivated in two ways. First, connections on E are identified with matrices, which act on vectors by multiplication and other matrices via the Lie bracket. The dB term is the vector part, i.e., the usual entry-wise derivative, and the $AB - BA$ term records the linear action of A on B.

The second way to motivate this is by decomposing $\mathrm{End}(E)$ as $\mathrm{End}(E) = E \otimes E^*$, and checking that tensoring ∇ with the connection on E^* obtained by dualizing gives the same thing.

Now that we have a connection on $\mathrm{End}(E)$, we again get a map $D_A : \Omega^0(\mathrm{End}\, E) \to \Omega^1(\mathrm{End}\, E)$, which we can also extend to a map $D_A : \Omega^i(\mathrm{End}\, E) \to \Omega^{i+1}(\mathrm{End}\, E)$.

Exercise 6
Use the Leibniz rule to extend D_A to a map $D_A : \Omega^i(\mathrm{End}\, E) \to \Omega^{i+1}(\mathrm{End}\, E)$.

Further, if A is a SU(2) connection, B is $\mathfrak{su}(2)$-valued, so $[A, B]$ is also $\mathfrak{su}(2)$-valued. Therefore we can restrict $\mathrm{End}(E)$ to $\mathrm{ad}(E)$ and set the complex

$$\Omega^0(\mathrm{ad}\, E) \xrightarrow{D_A} \Omega^1(\mathrm{ad}\, E) \to \cdots \qquad (*)$$

Lemma 7
For $B \in \Omega^1(\mathrm{ad}\, E)$, $D_A B$ as defined above is the $D_A B$ defined. !! Sign issue?

Proof. First, any $B \in \Omega^1(\mathrm{ad}\, E)$ can be expressed as a linear combination of 1-forms ω with coefficients constant $\mathfrak{su}(2)$ matrices C. Thus we can assume $B = \omega C$.

Then $D_A B = D_A(\omega C)$

$$\underset{\substack{\text{Leibniz} \\ \text{and } \omega \text{ is a 1-form}}}{=} d\omega\, C - \omega \wedge \nabla C$$

$$\underset{(\text{since } C \text{ const})}{=} d\omega\, C - \omega \wedge (dC + AC - CA)$$

$$\underset{\substack{(\text{since } \omega, A \text{ both} \\ \text{1-forms,} \\ \omega \wedge A = -A \wedge \omega)}}{=} d\omega\, C - \omega \wedge [A, C] = d\omega\, C - \underbrace{(\omega \wedge A)}\, C + \omega C \wedge A$$

$$= d\omega\, C - (-A \wedge \omega) C + \omega C \wedge A$$

$$= d(\omega C) + A \wedge (\omega C) + (\omega C) \wedge A$$

$$= dB + A \wedge B + B \wedge A.$$

□

Note that this proof, applied to $\omega = i$-form instead, shows in general that the map

$$D_A B : \Omega^i(\mathrm{End}\, E) \to \Omega^{i+1}(\mathrm{End}\, E) \text{ is}$$

$$D_A B = dB + A \wedge B - B \wedge A, \text{ for } i \text{ even},$$

$$\text{and } D_A B = dB + A \wedge B + B \wedge A, \text{ for } i \text{ odd}.$$

Now, seemingly from nowhere, we'll define the Bianchi Identity – this will be incredibly useful later.

Theorem 8 (Bianchi Identity)
If A is a fixed connection, $D_A F_A = 0$, $F_A \in \Omega^2(\mathrm{ad}(E))$.

Proof. Since $F_A = dA + A \wedge A$,

$$D_A F_A = dF_A + A \wedge F_A - F_A \wedge A$$

$$= d(dA + A \wedge A) + A \wedge (dA + A \wedge A) - (dA + A \wedge A) \wedge A$$

$$= \left(d^2 A + d(A \wedge A)\right) + \left(A \wedge dA + A \wedge A \wedge A\right) - \left(dA \wedge A + A \wedge A \wedge A\right)$$

$$= d^2 A + d(A \wedge A) + A \wedge dA + A \wedge A \wedge A - dA \wedge A - A \wedge A \wedge A$$

$$\underset{\substack{(\text{since } d^2A=0 \text{ and} \\ dA \wedge A = -A \wedge dA)}}{=} (dA \wedge A + A \wedge dA) + A \wedge dA - dA \wedge A \qquad \text{Sign issue?}$$

$$= d^2 A + dA \wedge A + A \wedge dA + A \wedge dA + A \wedge A \wedge A - dA \wedge A - A \wedge A \wedge A$$

$$= 0.$$

□

Now, let $D_A^* := \star D_A \star$. This gives us a codifferential

$$D_A^* : \Omega^2(\mathrm{ad}\, E) \to \Omega^1(\mathrm{ad}\, E).$$

We are now ready to return to discussing the critical points of the Yang-Mills Functional.

Yang-Mills Connections

We have seen that A is a critical point of YM iff

$$\langle D_A B, F_A \rangle_{L^2} = 0 \quad \text{for all } B \in \Omega^1(\mathrm{ad}\, E)$$

$$\Leftrightarrow \langle B, D_A^* F_A \rangle_{L^2} = 0$$

$$\Leftrightarrow D_A^* F_A = 0.$$

Definition 9
A connection such that $D_A^* F_A = 0$ is called a Yang-Mills connection. These connections are the critical points of the YM functional.

Note the similarity with Hodge theory! Last week, we saw that the critical points of L^2 norm on forms are precisely the harmonic forms: α for which $d\alpha = d^*\alpha = 0$.

If a connection A is Yang-Mills, then $D_A^* F_A = 0$. And by the Bianchi identity, we always have that $D_A F_A = 0$. Therefore YM connections can be thought of as harmonic forms; the slogan here is that "YM theory is the SU(2) version of Hodge theory."

The one caveat is that in general, $D_A^* F_A$ is difficult to understand. In Hodge theory, d always exists, but here, both D_A^* and F_A depend on A.

It turns out that some YM connections satisfy a simpler condition, which also guarantees that they are global minima/maxima.

Lemma 10 1. If A is SD or ASD, then $D_A^* F_A = 0$.

2. In fact, (A)SD are the global minima/maxima of the YM functional.

Proof.

1. This is immediate from Bianchi: since $D_A F_A = 0$ and $D_A^* = \star D_A \star$, if $F_A = \pm \star F_A$, then

$$D_A^* F_A = \star D_A \star F_A = \star D_A \pm F_A = \pm \star D_A F_A = 0.$$

2. For $F_A \in \Omega^2(\operatorname{ad} E) = \Omega^2_+(\operatorname{ad} E) \oplus \Omega^2_-(\operatorname{ad} E)$, it splits as $F_A = F_A^+ + F_A^-$, where $\langle F_A^+, F_A^- \rangle_{L^2} = 0$. Thus

$$\begin{aligned} \mathrm{YM}(\nabla) = \langle F_A, F_A \rangle_{L^2} &= -\int_M \operatorname{tr}(F_A \wedge \star F_A) \\ &= \langle F_A^+ + F_A^-,\ F_A^+ + F_A^- \rangle_{L^2} \\ &= \langle F_A^+, F_A^+ \rangle_{L^2} + \langle F_A^-, F_A^- \rangle_{L^2} + \langle F_A^+, F_A^- \rangle_{L^2} + \langle F_A^-, F_A^+ \rangle_{L^2} \\ &= \|F_A^+\|^2_{L^2} + \|F_A^-\|^2_{L^2}. \end{aligned}$$

Now, recall the instanton number k defined as

$$k = -\langle c_2, [M] \rangle = \frac{-1}{8\pi^2} \int_M \operatorname{tr}(F_A \wedge F_A).$$

Since $F_A = \star(\star F_A)$,

$$\begin{aligned} k &= \frac{1}{8\pi^2} \langle F_A, \star F_A \rangle_{L^2} \\ &= \frac{1}{8\pi^2} \langle F_A^+ + F_A^-,\ \star(F_A^+ + F_A^-) \rangle_{L^2} \\ &= \frac{1}{8\pi^2} \langle F_A^+ + F_A^-,\ F_A^+ - F_A^- \rangle_{L^2} \\ &= \frac{1}{8\pi^2} \Big(\langle F_A^+, F_A^+ \rangle_{L^2} + \langle F_A^+, -F_A^- \rangle_{L^2} + \langle F_A^-, F_A^+ \rangle_{L^2} + \langle F_A^-, -F_A^- \rangle_{L^2} \Big) \\ &= \frac{1}{8\pi^2} \Big(\|F_A^+\|^2_{L^2} - \|F_A^-\|^2_{L^2} \Big). \end{aligned}$$

Therefore $8\pi^2 k = \|F_A^+\|^2_{L^2} - \|F_A^-\|^2_{L^2}$. Combining this with $\mathrm{YM}(\nabla) = \|F_A^+\|^2_{L^2} + \|F_A^-\|^2_{L^2}$, we see that

if $k > 0$, then $\mathrm{YM} \geq 8\pi^2 k$, with equality when $F_A^- = 0$, i.e., when A is self-dual. On the

other hand, if $k < 0$, then $\mathrm{YM} \geq 8\pi^2(-k)$ and equality is realized when $F_A^+ = 0$, so A is anti-self-dual. $\square$